Functional Equations Problem Set 2

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Problem 1. (Centroamerican and Caribbean MO 2020 P3) Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(a) + f(b) + f(c) = a^2 + b^2 + c^2$$

for all $a, b, c \in \mathbb{Z}$ satisfying $a + b + c = 0$.

Problem 2. (Baltic Way 2025 P18) Find all functions $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $f(2) = 1$ and

$$\operatorname{lcm}(f(a+b), f(b)) \mid \operatorname{lcm}(a + f(b), b)$$

for all $a, b \in \mathbb{Z}_{>0}$.

Problem 3. (Pan American Girls' Mathematical Olympiad 2024 P5) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x+y) - f(x)) + f(x)f(y) = f(x^2) - f(x+y)$$

for all $x, y \in \mathbb{R}$.

Problem 4. (Rioplatense MO 2024 P5) Let $S = \{2, 3, 4, \dots\}$ be the set of positive integers greater than 1. Find all functions $f : S \to S$ such that

$$\gcd(a, f(b)) \cdot \operatorname{lcm}(f(a), b) = f(ab)$$

for all $a, b \in S$.

Problem 5. (Swiss MO 2020 P1) Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m) + f(n) \mid m + n$$

for all $m, n \in \mathbb{N}$.

Problem 6. (IMO 2022 P2) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for each $x \in \mathbb{R}^+$, there is exactly one $y \in \mathbb{R}^+$ such that

$$xf(y) + yf(x) \leq 2$$

Problem 7. (Romanian Masters of Mathematics Shortlist 2021 A1) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xy + f(x)) + f(y) = xf(y) + f(x + y)$$

for all $x, y \in \mathbb{R}$.

Problem 8. (IMEO 2019 P3) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$(x + y) \cdot f(x + y) = f(f(x) + y) \cdot f(x + f(y))$$

for all $x, y \in \mathbb{R}$.